Linux File System Explorer

1. Linux Directory: `/etc`

Item	Team response
AI Question	What is the purpose of `/etc`?
AI Answer	`/etc` stores system-wide configuration files, such as network settings, user account information, and service configurations.
Our Prediction	We expected `/etc` to contain configuration files used to manage Linux system settings.
Test	Run `ls /etc`
Actual Result	The command displayed configuration files and directories, such as `passwd`, `hosts`, and other system configuration files. Exact contents vary by system.
AI Correct?	Yes, if the observed contents match the explanation.
Final Learning	We learned that `/etc` is mainly used to store Linux system and service configuration files.

2. Linux Directory: `/var`

Item	Team response
AI Question	What is `/var` used for?

AI Answer	`/var` stores variable data that changes during system operation, including log files, caches, and application data.
Our Prediction	We expected `/var` to contain log files and other data that changes frequently.
Test	Run `ls /var` and `ls /var/log`
Actual Result	The commands displayed variable-data directories and, where present and accessible, system log files.
AI Correct?	Yes, if the observed contents match the explanation.
Final Learning	We learned that `/var` stores changing system and application data. `/var/log` is a common location for investigating logs.

3. Linux Directory: `/home`

Item	Team response
AI Question	What is `/home` used for?
AI Answer	`/home` contains personal directories for regular users, where they can store documents, downloads, and personal configuration files.
Our Prediction	We expected `/home` to contain directories belonging to individual users.

Test	Run `ls /home`
Actual Result	The command displayed user home directories available on the system. The contents depend on the configured users and permissions.
AI Correct?	Yes, if the observed contents match the explanation.
Final Learning	We learned that `/home` is the standard location for regular users' personal files and settings.

4. Linux Directory: `/tmp`

Item	Team response
AI Question	What is `/tmp` used for?
AI Answer	`/tmp` stores temporary files created by applications and users. These files may be deleted automatically during a reboot or by system cleanup policies.
Our Prediction	We expected `/tmp` to contain temporary files that applications use while running.
Test	Run `ls -la /tmp`
Actual Result	The command displayed files and directories in `/tmp`, including hidden entries if present. The exact contents depend on current system activity.

AI Correct?	Yes, if the observed contents and system behavior are consistent with the explanation.
Final Learning	We learned that `/tmp` is intended for temporary data and that we should not assume files stored there will remain permanently.

5. Linux Directory: `/usr`

Item	Team response
AI Question	What is `/usr` used for?
AI Answer	`/usr` contains many user-space programs, shared libraries, documentation, and other resources used by applications and the operating system.
Our Prediction	We expected `/usr` to contain application commands, libraries, and supporting files.
Test	Run `ls /usr` and `ls /usr/bin`
Actual Result	The commands displayed directories such as `bin`, `lib`, `share`, and many installed command-line programs, depending on the Linux distribution.
AI Correct?	Yes, if the observed contents match the explanation.
Final Learning	We learned that `/usr` contains many installed programs and shared resources needed by Linux applications and users.

Linux Commands Investigation

1. Linux Command: `pwd`

Item	Team response
AI Question	What does `pwd` do?
AI Answer	`pwd` stands for Print Working Directory. It displays the full path of the directory you are currently working in.
Our Prediction	We expected the command to show our current location in the Linux file system.
Test	Run `pwd`
Actual Result	The terminal displays the current directory path, for example, `/home/user`.
AI Correct?	Yes, if the output shows the current working directory.
Final Learning	We learned that `pwd` helps us identify our current location in the Linux file system.

2. Linux Command: `ls` vs. `ls -la`

Item	Team response
AI Question	What is the difference between `ls` and `ls -la`?
AI Answer	`ls` lists directory contents, usually excluding hidden files. `ls -la` lists all entries, including hidden files, in long format with details such as permissions, ownership, size, and modification time.

Our Prediction	We expected `ls` to display a simple list and `ls -la` to display more detailed information, including hidden files.
Test	Run `ls` and then `ls -la`
Actual Result	`ls` displays a basic list of entries. `ls -la` displays detailed information and includes hidden entries whose names begin with a dot (`.`).
AI Correct?	Yes, if the observed output matches the explanation.
Final Learning	We learned that `ls` is useful for quickly viewing directory contents, while `ls -la` provides more detailed information and shows hidden entries.

3. Linux Command: `cd`

Item	Team response
AI Question	What does `cd` do?
AI Answer	`cd` stands for Change Directory. It changes the current working directory in the terminal.
Our Prediction	We expected the command to move us from one directory to another.
Test	Run `cd /tmp`, followed by `pwd`.
Actual Result	The terminal changes to `/tmp`, and `pwd` displays `/tmp` as the current directory.

AI Correct?	Yes, if the current directory changes as expected.
Final Learning	We learned that `cd` helps us navigate between directories in Linux.

4. Linux Command: `mkdir`

Item	Team response
AI Question	What does `mkdir` do?
AI Answer	`mkdir` stands for Make Directory. It creates a new directory.
Our Prediction	We expected the command to create a new folder at the specified location.
Test	Run `mkdir practice` and then `ls -ld practice`.
Actual Result	A directory named `practice` is created in the current directory, provided it does not already exist and we have the required permissions.
AI Correct?	Yes, if the directory is created successfully.
Final Learning	We learned that `mkdir` is used to organize files by creating new directories.

5. Linux Command: `rm`

Item	Team response

AI Question	What does `rm` do?
AI Answer	`rm` stands for Remove. It deletes files. With appropriate options, it can also remove directories and their contents.
Our Prediction	We expected the command to delete a specified file from the file system.
Test	Run `touch testfile.txt`, followed by `rm testfile.txt`, then `ls -l testfile.txt`.
Actual Result	The file is deleted. The final `ls` command reports that the file cannot be found.
AI Correct?	Yes, if the file is removed successfully.
Final Learning	We learned that `rm` deletes files and must be used carefully because deleted data may be difficult to recover.

Important safety note: Avoid experimenting with `rm -rf` on system directories or important files. It can recursively delete large amounts of data without prompting for confirmation.

Linux Permissions

1. Meaning of `rwx`

Item	Team response
AI Question	What does `rwx` mean in Linux permissions?

AI Answer	`rwx` stands for read (r), write (w), and execute (x). Read allows viewing file contents, write allows modifying a file, and execute allows running a file as a program or script. For directories, these permissions have different meanings: read lists entries, write allows creating or removing entries (subject to other restrictions), and execute allows accessing entries and traversing the directory.
Our Prediction	We expected `rwx` to represent the three basic permissions that control access to files and directories.
Test	Run `ls -l` to view permissions on files and directories.
Actual Result	The output may show `-rwxr-xr--` or similar permission strings. The first character identifies the file type; the next nine characters represent permissions for owner, group, and others.
AI Correct?	Yes, if the observed permissions match the explanation.
Final Learning	We learned that `rwx` represents read, write, and execute permissions, and that their effects depend on whether the target is a file or directory.

2. Owners, Groups and Others

Item	**Team response**
AI Question	What are owner, group, and others in Linux permissions?
AI Answer	Owner is the user who owns the file. Group refers to users in the file's associated group. Others refers to everyone else. Linux assigns permission bits separately to these three categories.

Our Prediction	We expected Linux to separate access permissions based on the user who owns a file, the associated group, and all other users.
Test	Run `ls -l filename` on an existing file, replacing `filename` with its name.
Actual Result	The output displays the file's owner, group, and permission string, subject to the user's access to the file's directory.
AI Correct?	Yes, if the output and file ownership information match the explanation.
Final Learning	We learned that Linux permissions allow different levels of access for the owner, group members, and other users.

3. `chmod 755`

Item	Team response
AI Question	What does `chmod 755` mean?
AI Answer	`chmod` changes file permissions. In `755`, the owner has read, write, and execute permissions (`rwx`); the group and others have read and execute permissions (`r-x`).
Our Prediction	We expected the owner to have full permissions while the group and others could read and execute the file but not modify it.
Test	Run `chmod 755 script.sh` and then `ls -l script.sh` on a test file you own.

Actual Result	The permission string should show `-rwxr-xr-x` for a regular file, assuming the command succeeds and no special permission bits are involved.
AI Correct?	Yes, if the displayed permission bits match the explanation.
Final Learning	We learned that 755 grants full permissions to the owner and read-and-execute permissions to the group and others. It is commonly used for executable scripts and directories.

4. `chmod 644`

Item	Team response
AI Question	What does `chmod 644` mean?
AI Answer	In `644`, the owner has read and write permissions (`rw-`), while the group and others have read-only permissions (`r--`). No category has execute permission.
Our Prediction	We expected the owner to be able to read and modify the file, while everyone else could only read it.
Test	Run `chmod 644 document.txt` and then `ls -l document.txt` on a test file you own.
Actual Result	The permission string should show `-rw-r--r--` for a regular file, assuming the command succeeds and no special permission bits are involved.
AI Correct?	Yes, if the displayed permission bits match the explanation.

Final Learning	We learned that 644 is commonly used for regular files that the owner can edit and other users can read, without granting execute permission.

5. Disadvantages of `chmod 777`

Item	Team response
AI Question	Why is `chmod 777` generally discouraged?
AI Answer	`chmod 777` grants read, write, and execute permissions to the owner, group, and others. This can allow other users to modify or execute files, creating security risks such as unauthorized changes, malicious code execution, or data loss.
Our Prediction	We expected 777 to give everyone full permissions and potentially expose files to misuse.
Test	On a disposable test file you own, run `chmod 777 testfile.txt` and then `ls -l testfile.txt`. Avoid using this on important files or system directories.
Actual Result	The permission string should show `-rwxrwxrwx` for a regular file, assuming the command succeeds and no special permission bits are involved.
AI Correct?	Yes, if the resulting permissions match the explanation.
Final Learning	We learned that 777 gives every user full basic permissions. It should generally be avoided because it violates the principle of least privilege. Grant only the permissions users actually need.

The numbers are calculated as `r = 4`, `w = 2`, and `x = 1`, adding the values for the permissions granted to each category.

- 755 = `rwxr-xr-x`
- 644 = `rw-r--r--`
- 777 = `rwxrwxrwx`

Linux Processes:

1. Linux process

Item	Team response
AI Question	What is a Linux process?
AI Answer	A process is a running instance of a program. Each process has a unique Process ID (PID).
Our Prediction	We expected every running program to have a PID that identifies it.
Test	Run `sleep 100 &` and then `ps`.
Actual Result	The `sleep` process runs in the background and appears with its PID.
AI Correct?	Yes, if the observed output matches the explanation.
Final Learning	We learned that a process is a running program identified by its PID.

2. ps Command

Item	Team response
AI Question	What does ps show?
AI Answer	The ps command displays information about processes, typically those associated with the current terminal.
Our Prediction	We expected ps to display the processes running on the system.
Test	Run ps.
Actual Result	The command displayed a limited list of processes associated with the current terminal session.
AI Correct?	Yes, if the observed output matches the explanation.
Final Learning	We learned that ps shows a snapshot of selected processes, not necessarily every process on the system.

3. ps aux Command

Item	Team response
AI Question	What does `ps aux` show?
AI Answer	`ps aux` displays detailed information about processes across the system, including users, PIDs, CPU usage, and memory usage.
Our Prediction	We expected `ps aux` to show more processes and details than `ps`.
Test	Run `ps aux`.
Actual Result	The command displayed a detailed list of system processes and their resource usage.
AI Correct?	Yes, if the observed output matches the explanation.
Final Learning	We learned that `ps aux` helps inspect processes belonging to different users and identify resource usage.

4. `top` Command

Item	Team response

AI Question	What does `top` do?
AI Answer	`top` displays a continuously updating view of running processes and system resource usage, including CPU and memory.
Our Prediction	We expected `top` to show process information that updates periodically.
Test	Run `top` and press `q` to exit.
Actual Result	The command displayed running processes and periodically refreshed system and resource statistics.
AI Correct?	Yes, if the observed output matches the explanation.
Final Learning	We learned that `top` is useful for monitoring system performance and identifying processes consuming excessive resources.

5. Why would a developer investigate processes?

Item	Team response
AI Question	Why would a developer investigate processes?

AI Answer	Developers investigate processes to troubleshoot slow applications, high CPU or memory usage, frozen programs, and unexpected background activity.
Our Prediction	We expected process inspection to help identify applications causing performance problems.
Test	Run `sleep 100 &`, then use `ps aux` and `top` to locate the process.
Actual Result	The test process appeared in the process list, and its resource usage could be inspected.
AI Correct?	Yes, if the observed output matches the explanation.
Final Learning	We learned that process monitoring helps developers troubleshoot performance issues and understand what is running on a Linux system.

1. Linux File System Explorer: Which directory should you investigate first when an application generates log files, and why?

Answer: `/var/log`, because it commonly stores system and application logs that help identify errors and troubleshoot problems.

2. Linux Commands Investigation: What happens if you execute `rm -rf`? Why is it dangerous?

Answer: `rm -rf` forcefully deletes files and directories recursively without prompting for confirmation. It is dangerous because incorrect use can cause significant data loss.

3. Linux Permissions: Is `chmod 777` always required when a user gets "Permission denied"?

Answer: No. Check file ownership and permissions first, then grant only the required permissions. `chmod 777` is discouraged because it gives everyone read, write, and execute permissions.

4. Linux Processes: If an application is running slowly, can `top` help investigate the problem?

Answer: Yes. `top` displays CPU usage, memory usage, process IDs (PIDs), and process names. This helps identify resource-intensive processes and investigate performance problems.